



Babel

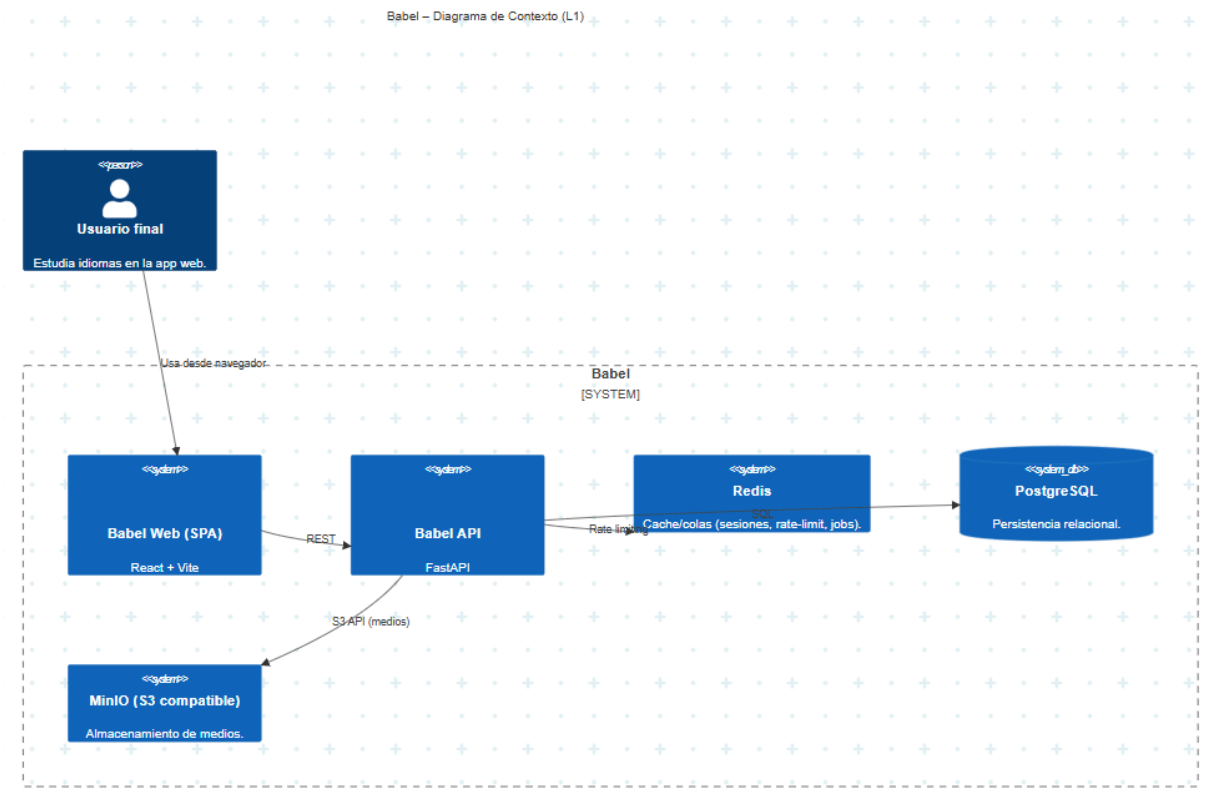
Victor Amaury Cespedes Linares

**Universidad Sergio Arboleda
Escuela de ciencias exactas e ingeniería**

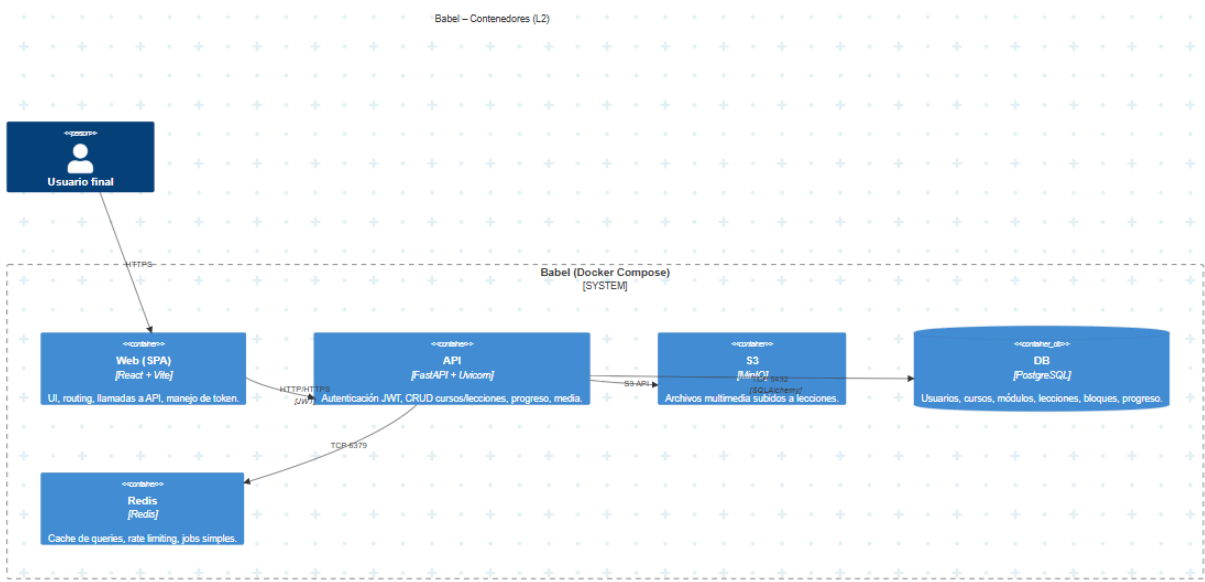
Agosto de 2025

DIAGRAMA C4

L1:

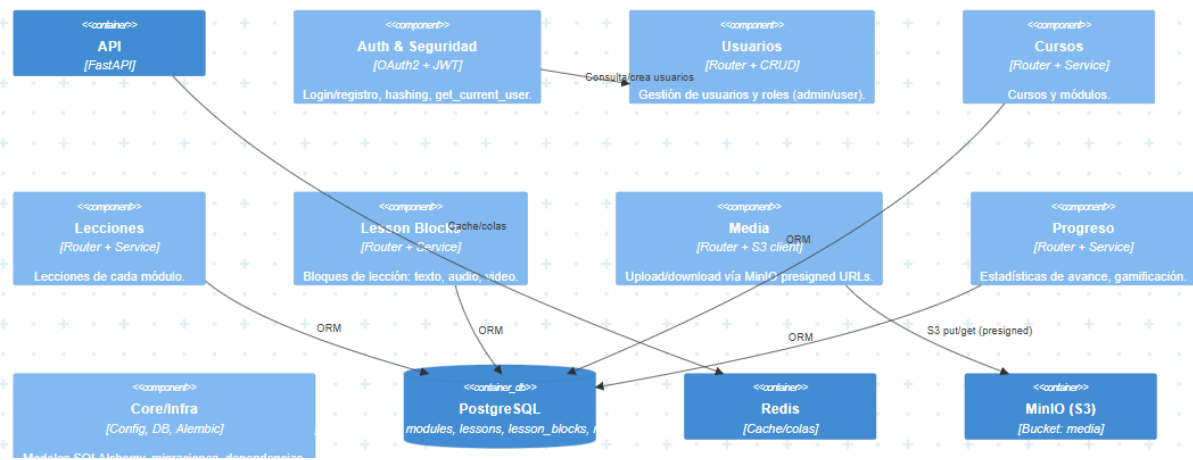


L2:

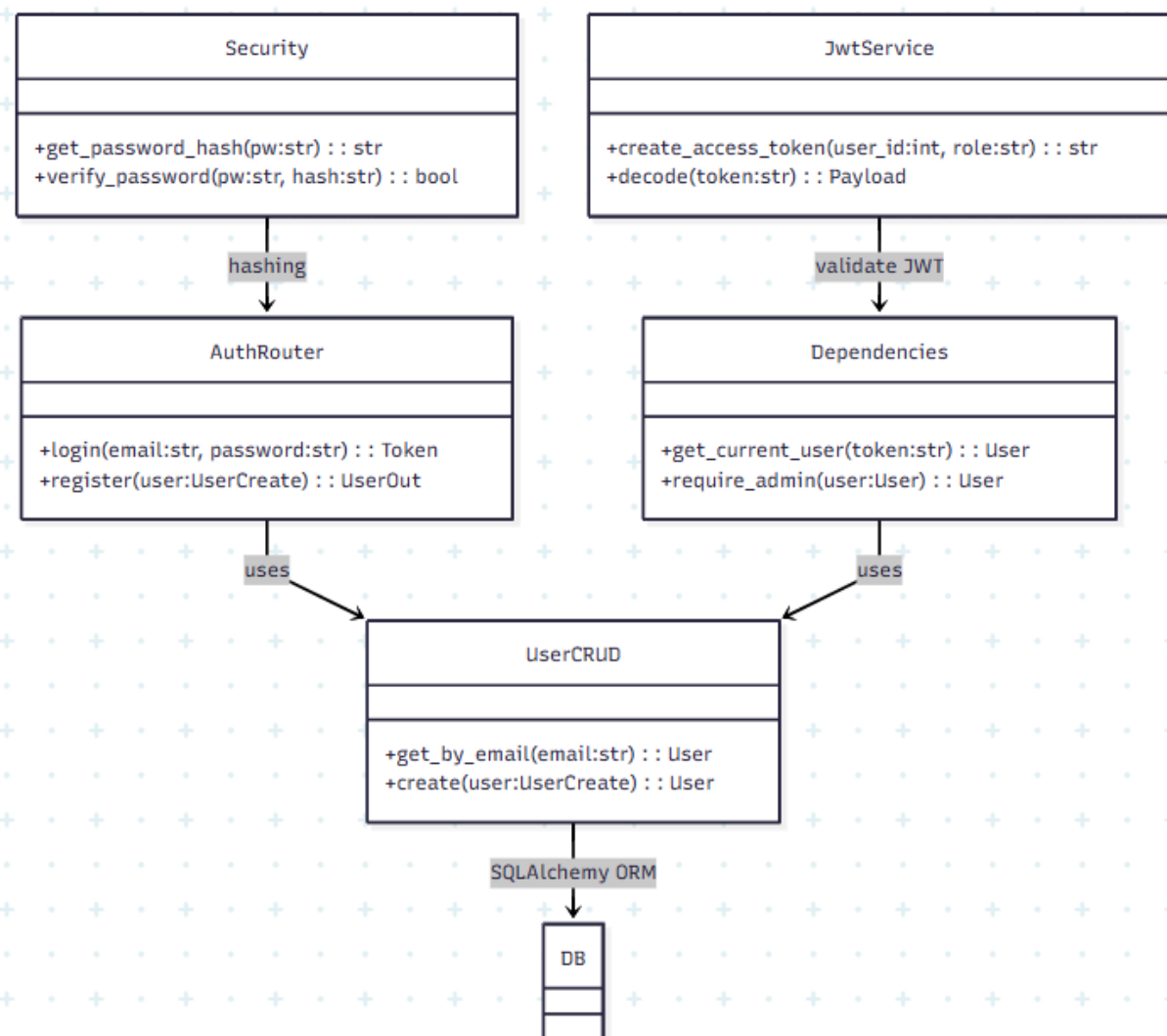


L3:

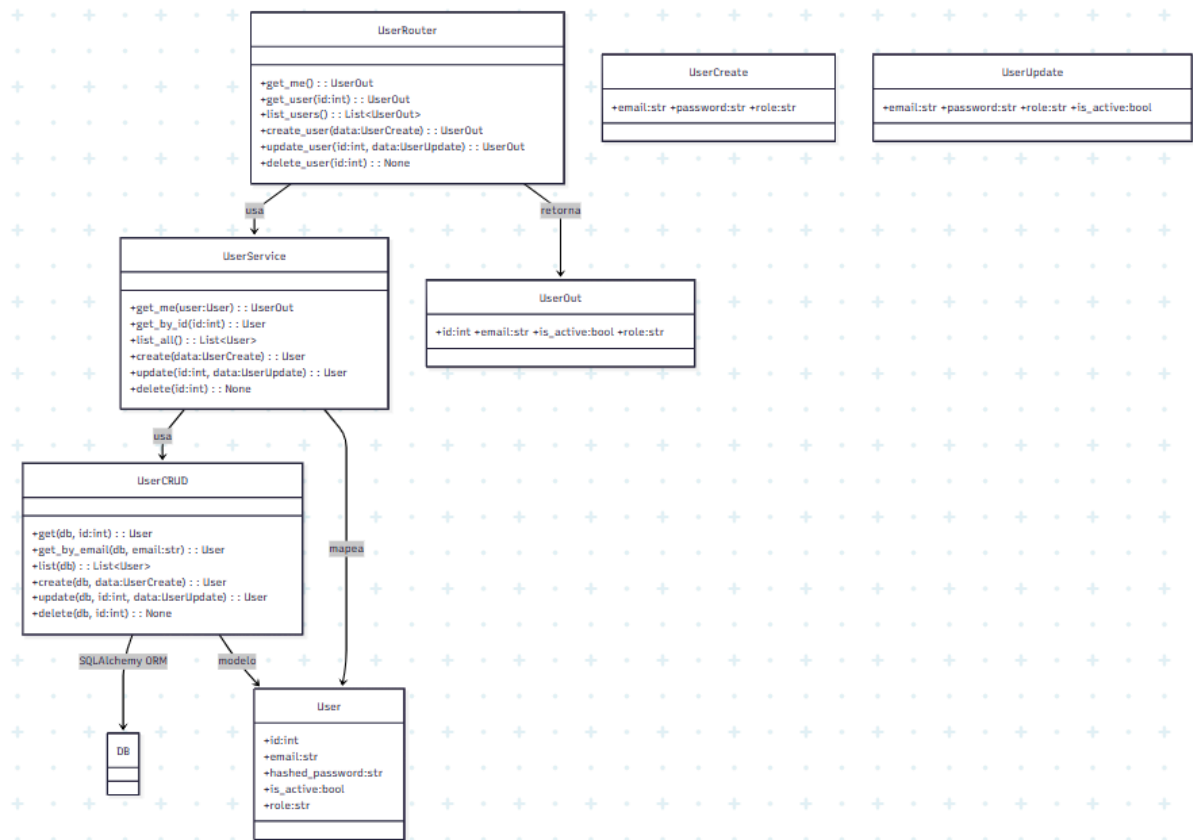
Babel API – Componentes (L3)



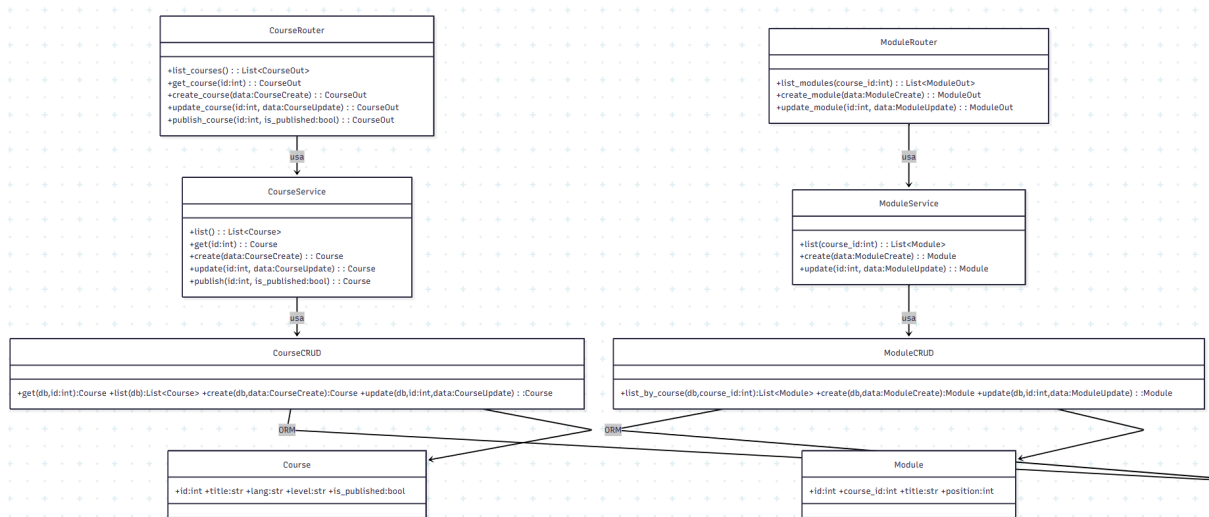
L4 Auth:

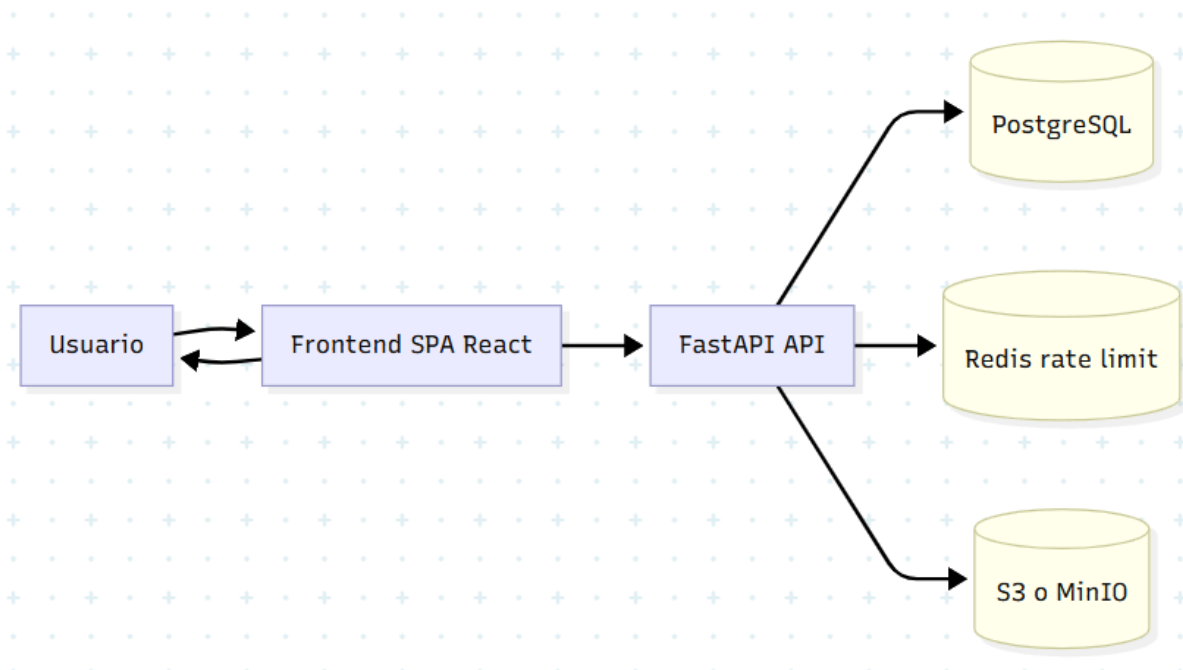


L4 Users:

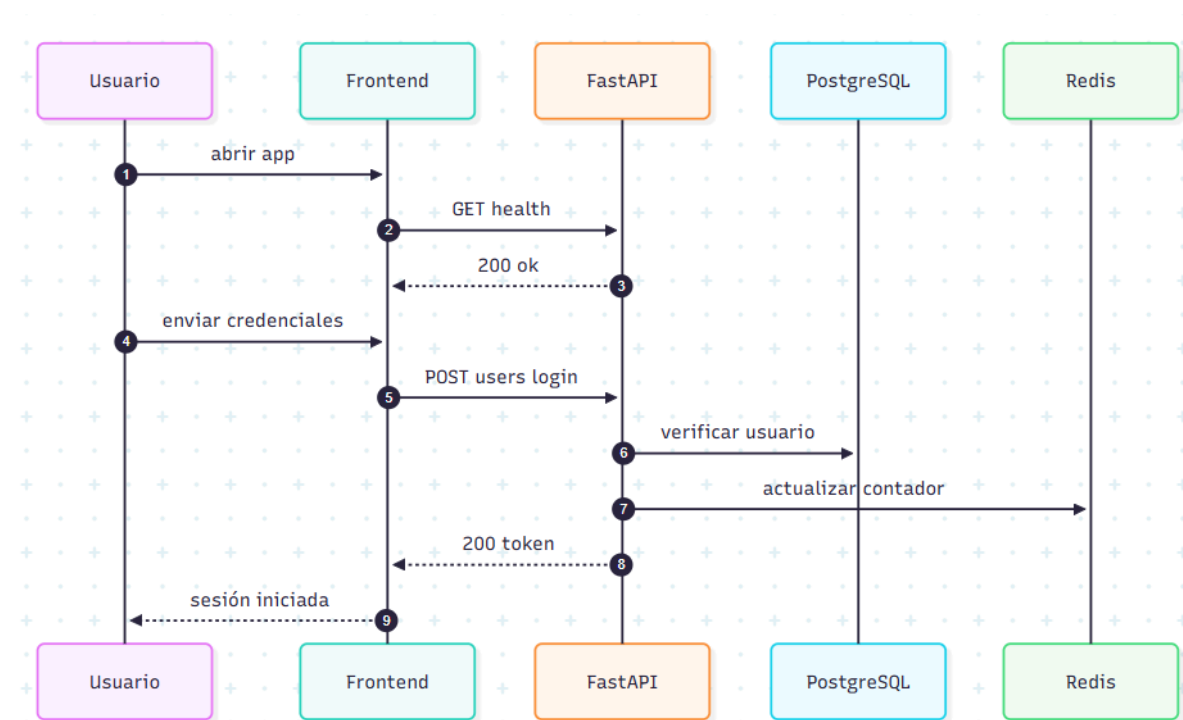


L4 Courses:





Login:



Cursos:

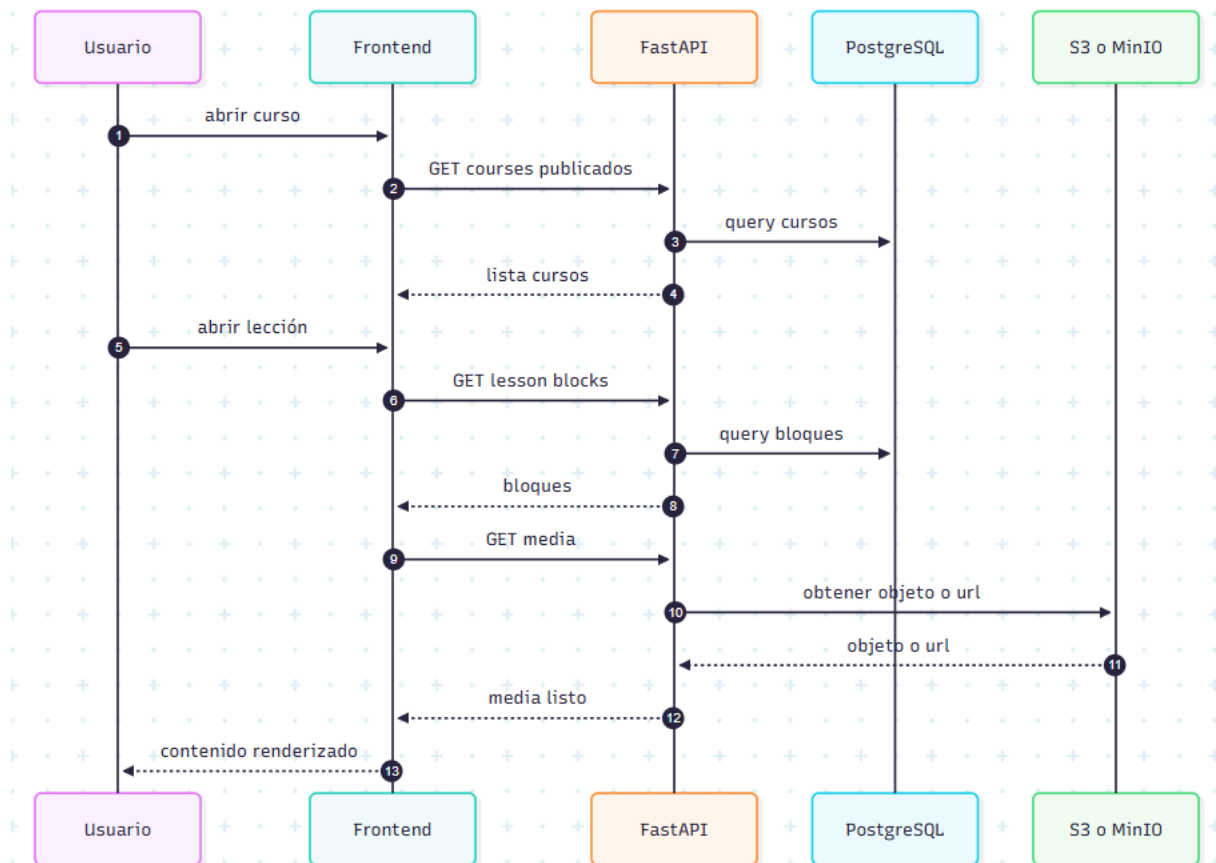
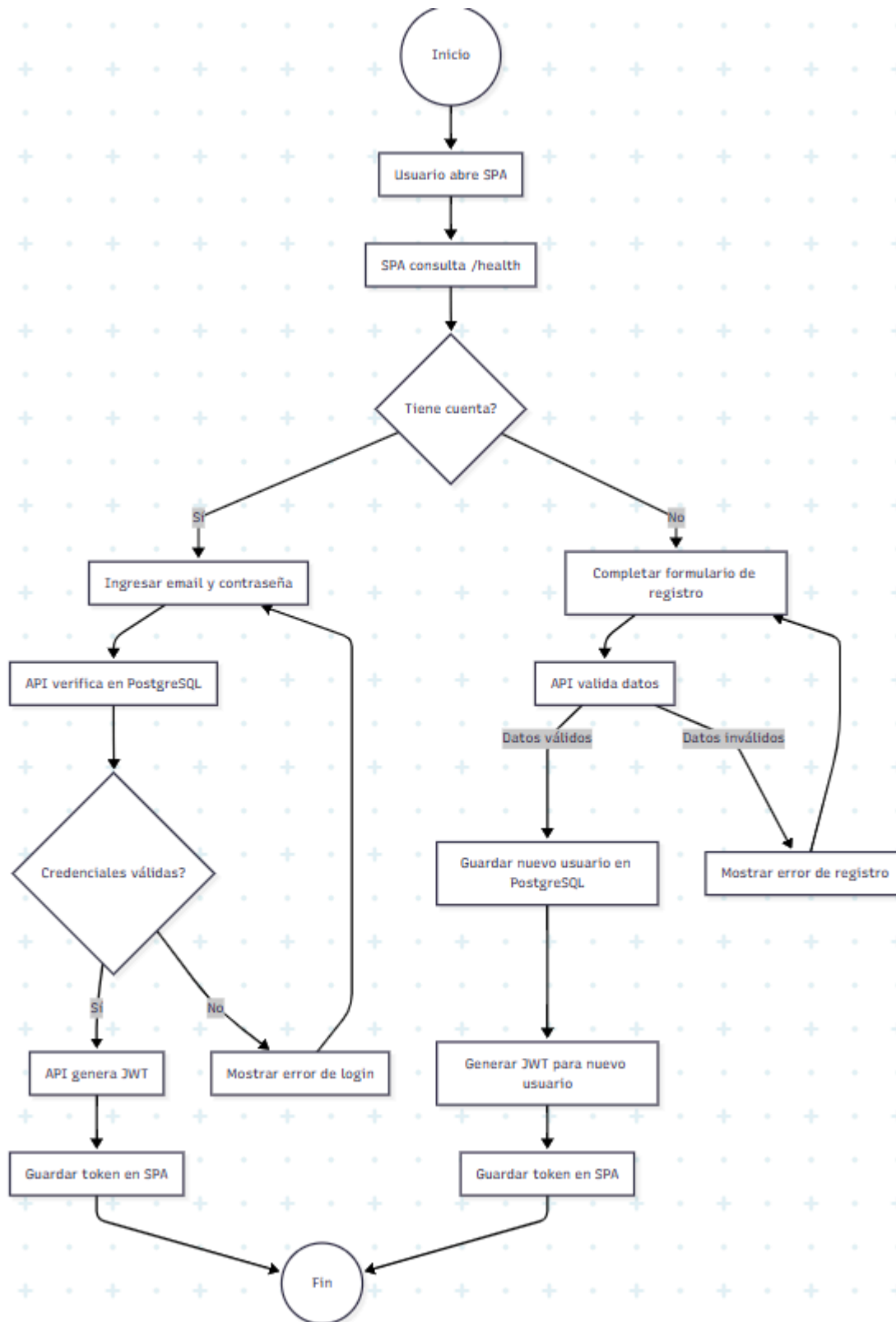
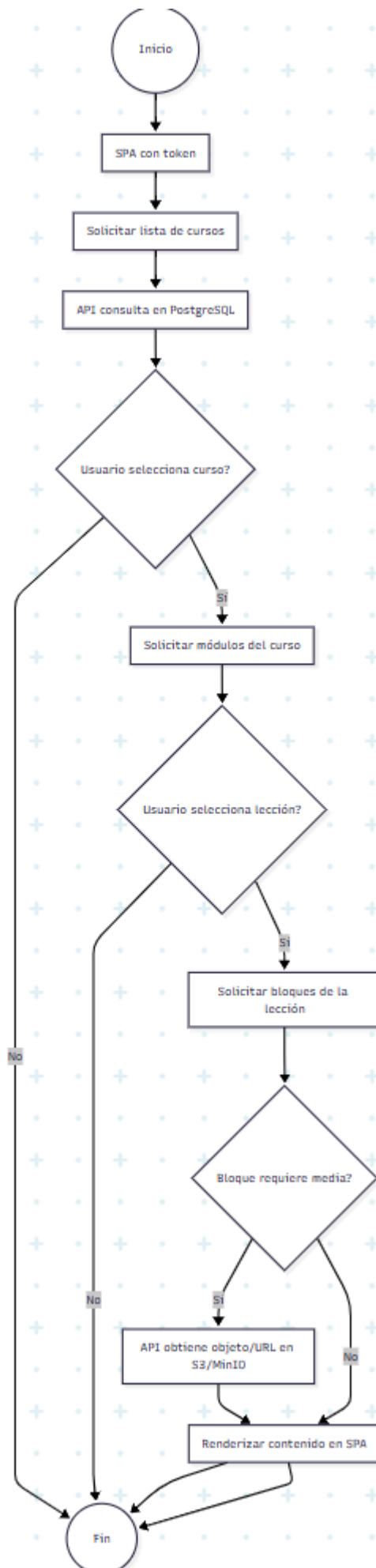


Diagrama de Flujo:

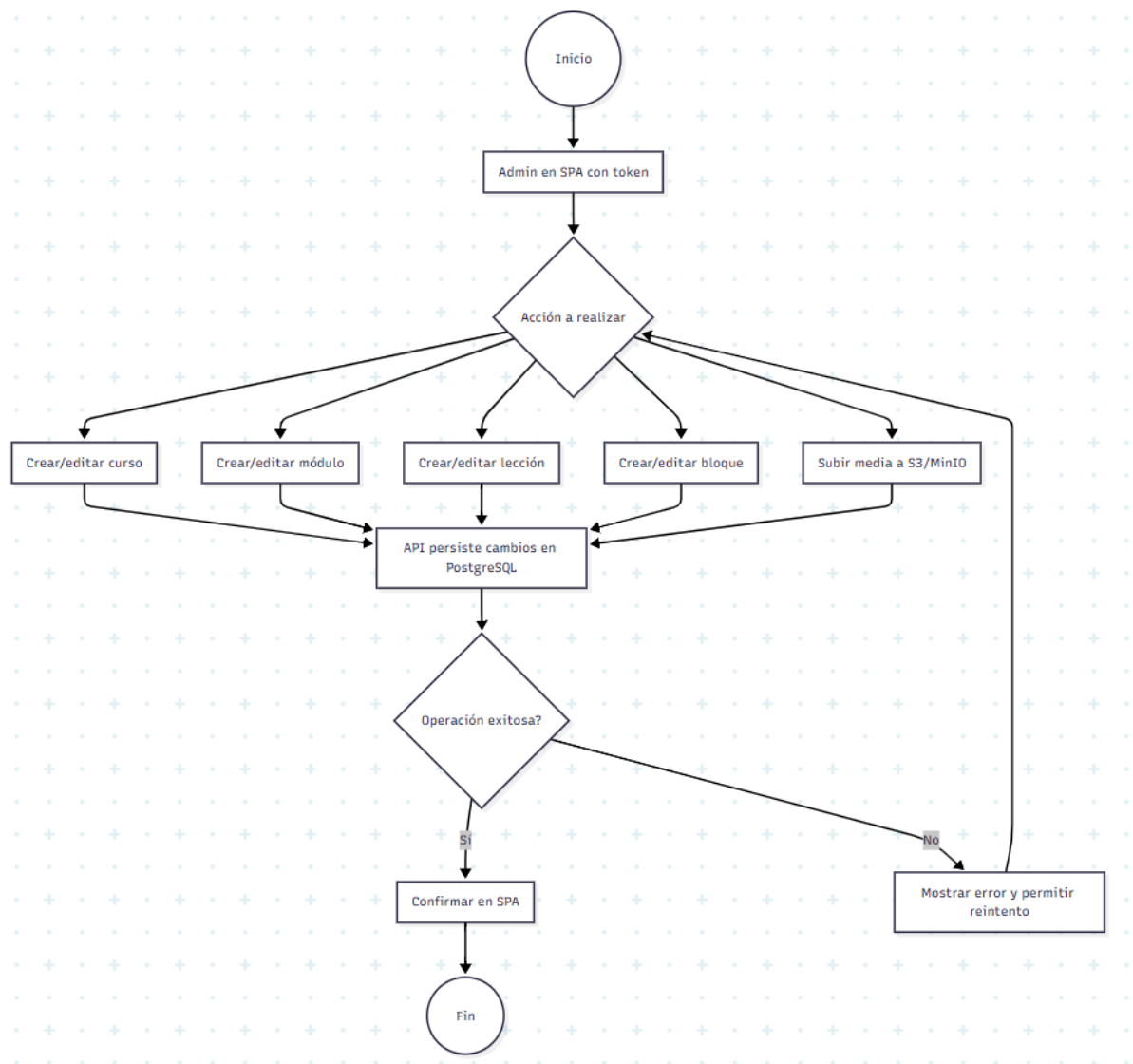
Inicio:





Consumo:

Admin

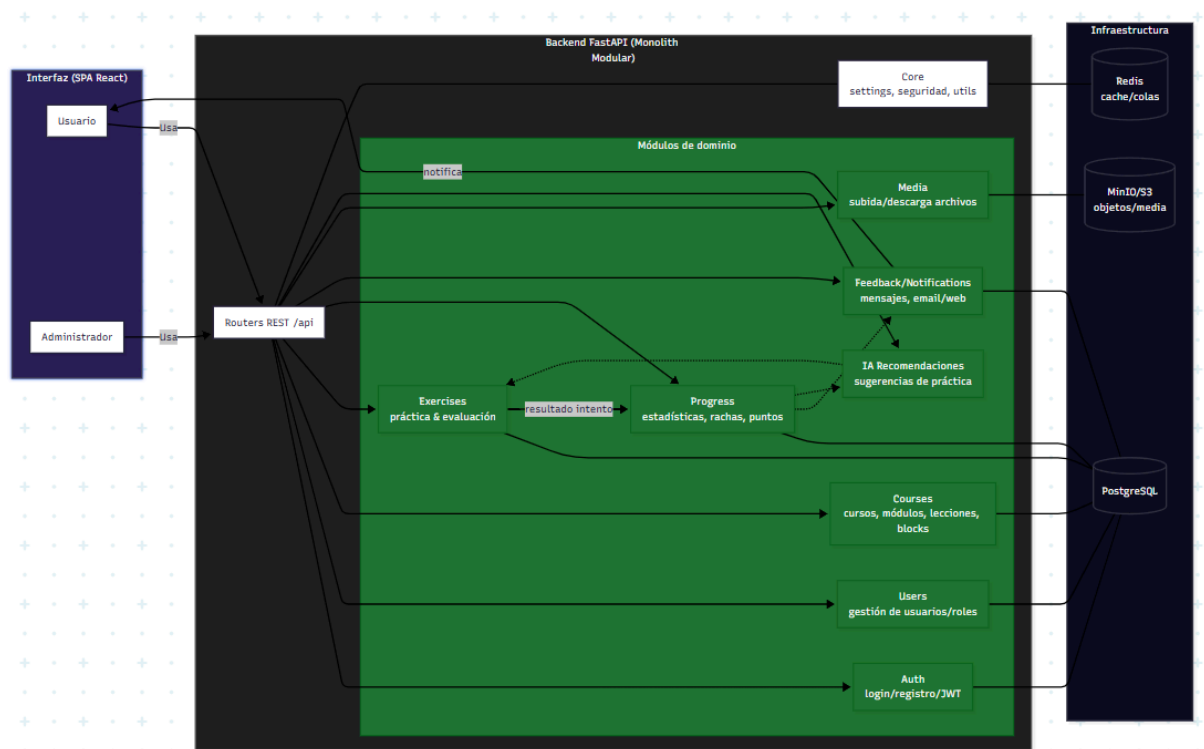


Arquitectura y Diseño

1. Patrones:

- a. Modular Monolith ---- Backend
- b. SPA ---- Frontend
- c. Rest ---- Comunicación

2. Diagrama Componentes:



3. Principios:

- SOLID
- KISS
- State
- Observer
- Lazy
- Adapter
- Least Privilege

Arquitectura Fisica y Logica

- División del sistema por modulos
- Interacción por eventos y consumo interno
- Ec2
 - t3.small
 - Contenedor back,front,bd,ia,redis,s3
 - S3, backups
- Escalabilidad
 - ALB --- Duplicar la app
 - 1 sólo s3
 - 1 sola db
 - 1 solo redis
- Seguridad

- a. Security groups Ec2
 - b. JWT
 - c. Rate_limit
 - d. Bcrypt
 - e. Endpoints
 - f. Time-outs
 - g. Token
 - h. Backups
 - i. .env
6. Desempeño
- a. Lazy load
 - b. Cache
 - c. Bd